Code No: 157DW **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
SOFT COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART – A

(25 Marks)

1.a)	List the differences between soft computing and hard computing.	[2]
b)	What is a neural network?	[3]
c)	What is relational join?	[2]
d)	State extension principle for fuzzy sets.	[3]
e)	Quote an example for multi criteria decision making.	[2]
f)	List the variants of particle swarm optimization.	[3]
g)	What is evolutionary computing?	[2]
h)	Illustrate multipoint crossover.	[3]
i)	How to reduce attributes in rough sets?	[2]
j)	List neuro fuzzy systems.	[3]

PART – B

(50 Marks)

2. Outline the recent trends and advancements in soft computing. [10]

OR

3. Explain in detail about various soft computing methods. [10]

4. Explain the properties of intuitionistic fuzzy relations and composition of two IFRs with examples. [10]

OR

5. With suitable examples, discuss fuzzy propositions used in fuzzy rule based systems. [10]

6. Compare and contrast problem solving with decision making. [10]

OR

7. Describe the clustering model of particle swarm optimization. [10]

8. Explain the basic concepts and operators of genetic algorithms. [10]

OR

9.a) Discuss the importance of fitness function in genetic algorithm.
b) Differentiate between mutation and crossover operations. [5+5]

10. Demonstrate fuzzy C-Means clustering algorithm. [10]

OR

11. Elaborate integration of soft computing techniques in information retrieval and semantic web. [10]

---ooOoo---